

Melody Slot Machine on iPhone

Dial-type Interface for Morphed Melody

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ABSTRACT

We present Melody Slot Machine on iPhone, an iPhone application using the melodic morphing method on the basis of the Generative Theory of Tonal Music (GTTM). We previously developed a demonstration system called Melody Slot Machine to introduce the melodic morphing method and presented it at international conferences and exhibitions. Since in-person demonstrations were reduced due to the Covid-19 Pandemic, we implemented an application with the same functionality and made it available for downloading to experience it. Our Melody Slot Machine on iPhone currently has two contents available for download, and we plan to add more contents in the future.

CCS CONCEPTS

• Applied computing → Sound and music computing.

KEYWORDS

generative theory of tonal music (gttm), melody morphing, slot machine, dial type interface

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1 INTRODUCTION

We have developed Melody Slot Machine on iPhone, an iPhone application that generates new combinations of melodies by manipulating a slot machine-like interface¹. The melody variations used in the Melody Slot Machine on iPhone are generated by a melodic morphing method on the basis of the Generative Theory of Tonal Music (GTTM) [15]. The melodic morphing method uses the musical structure extracted by using GTTM, and the original melodies used for morphing and the melodies generated by morphing have similar structures [6, 12]. Therefore, the global structure of multiple melodies generated by morphing does not change when a portion of one melody time axis is replaced by another melody.

¹<https://gttm.jp/hamanaka/en/melodyslotmachine/>

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The Melody Slot Machine, a research demonstration device, was developed to promote the melodic morphing method, and the performer's movements can be viewed on Pepper's ghost display (Fig. 1(a))². In the Melody Slot Machine, melody segments are displayed on a dial, and the melody to be played can be switched by rotating the dial (Fig. 1(b)). After the Melody Slot Machine was developed, it was exhibited at an international conference, but the Covid-19 Pandemic made it difficult to demonstrate Melody Slot Machine in person [4, 13, 16]. Therefore, we developed the Melody Slot Machine on iPhone with the aim of enabling many people to experience it by downloading an application.

(a) Pepper's ghost display



(b) Slot dial

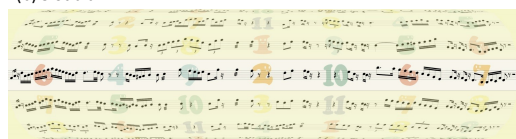


Figure 1: Melody Slot Machine

2 MELODY SLOT MACHINE ON IPHONE

Figure 2 is a screenshot of the Melody Slot Machine on iPhone. The horizontal axis is time, and each dial displays a melody segment in musical notation (Fig. 2(a)). By swiping up and down on each dial, you can switch the segments. Due to the limited screen size of the iPhone, only four melody segments can be viewed simultaneously, and the currently playing segment can be viewed by automatically scrolling left as the musical piece progresses (Fig. 2(b)).

The dial changes for the entire musical piece are displayed in a grid at the bottom of the screen, corresponding to the numbers

²<https://gttm.jp/hamanaka/en/melody-slot-machine/>

written on the dials. Swiping up from the bottom of the grid display brings up the full-screen grid (Fig. 2(c)). Touching the full-screen grid changes the selected grid. The change in the grid is linked to the dial, and the melody is played reflecting the change. Swiping down terminates the full-screen grid, and the dial appears again. When the iPhone is shaken up and down, each dial is shuffled to generate a new combination of melodies (Fig. 2(d)).

Melody Slot Machine on iPhone can generate a huge number of combinations of melodies. For example, in Fig. 2(c), 11 variation melodies are divided into 7 melody segments in the time direction, so there are $11^7 = 19,487,171$ combinations of melodies. Therefore, the user can feel as if he or she is composing a melody, because a combination of melodies that the user has not yet heard is generated by his/her own operation.

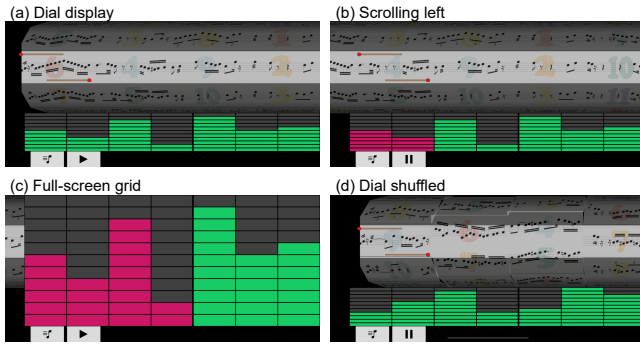


Figure 2: Screenshot of Melody Slot Machine on iPhone

3 MELODIC MORPHING METHOD

A melodic morphing method has been proposed that applies the reduction operation to generate a melody that is structurally intermediate between two input melodies by combining two melodies after reducing their time-span trees (Fig. 3)[12].

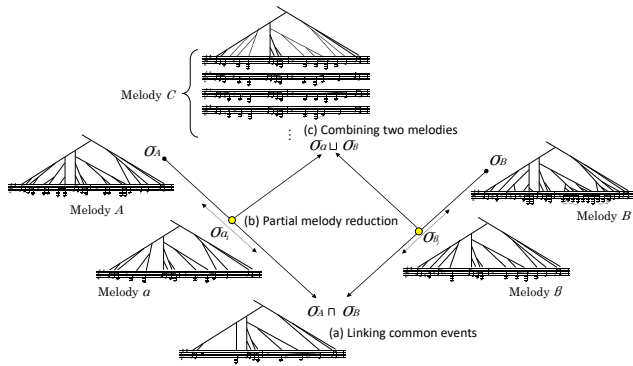


Figure 3: Melodic morphing method

4 RELATED WORK

The AI-based melody generation system has dramatically improved in quality compared with previous work, but the musical perfection

is not yet sufficient and needs to be reworked [1, 2, 14, 17]. Here, we consider the consistency and operability of music AI. In general, music created using learning-based approaches is locally consistent but lacks a global structure [3]. For example, although a relatively large level of structure can be generated by giving appropriate prompts in MusicLM, the desired music is difficult to generate in one trial, and trial and error of prompts is necessary to reflect the user's intentions.

In the trial-and-error process of production, users need to be able to perform appropriate operations that reflect their intentions and have high operability. We have studied GTTM as a highly consistent and high operability method.

At the start of app development, because of the low accuracy of the time-span tree analyzer of the GTTM, composers manually performed time-span tree analysis to generate morphing melodies [5]. The introduction of deep learning has dramatically improved the analysis performance [8–10], making it possible to automate from time-span tree analysis to morphing melody generation [11].

5 EXPERIMENTAL RESULTS

Melody Slot Machine on iPhone was launched in May 2021 and had been downloaded 657 times by July 2023. After launching the application 10 times, users are presented with a message inviting them to complete a questionnaire regarding the usability of the application. The questionnaire form can be displayed by following the instructions or by following the link from the app's settings page. The following is a portion of the questionnaire.

Q1: Which was easier to operate, the dial screen or the grid screen?

Q2: Which was more enjoyable to operate, the dial screen or the grid screen?

Thirty responses were received, and Table 1 shows the results. Many said that the dial type was easier to operate and more enjoyable than the grid type.

Table 1: Results of Questionnaire

	Dial screen	Grid screen	Not using grid screen
Q1	16	10	4
Q2	17	9	4

6 CONCLUSION

This paper describes an application called Melody Slot Machine on iPhone, which uses a melodic morphing method on the basis of the Generative Theory of Tonal Music (GTTM). Currently, two musical pieces have been released, and we plan to add more content in the future.

The morphing described in Section 3 was intended for melodies that are interpolations of two melodies. However, we also proposed a method for generating extrapolated melodies [7]. In the future, we will realize automatic generation of extrapolated melodies and develop applications that use extrapolated melodies.

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